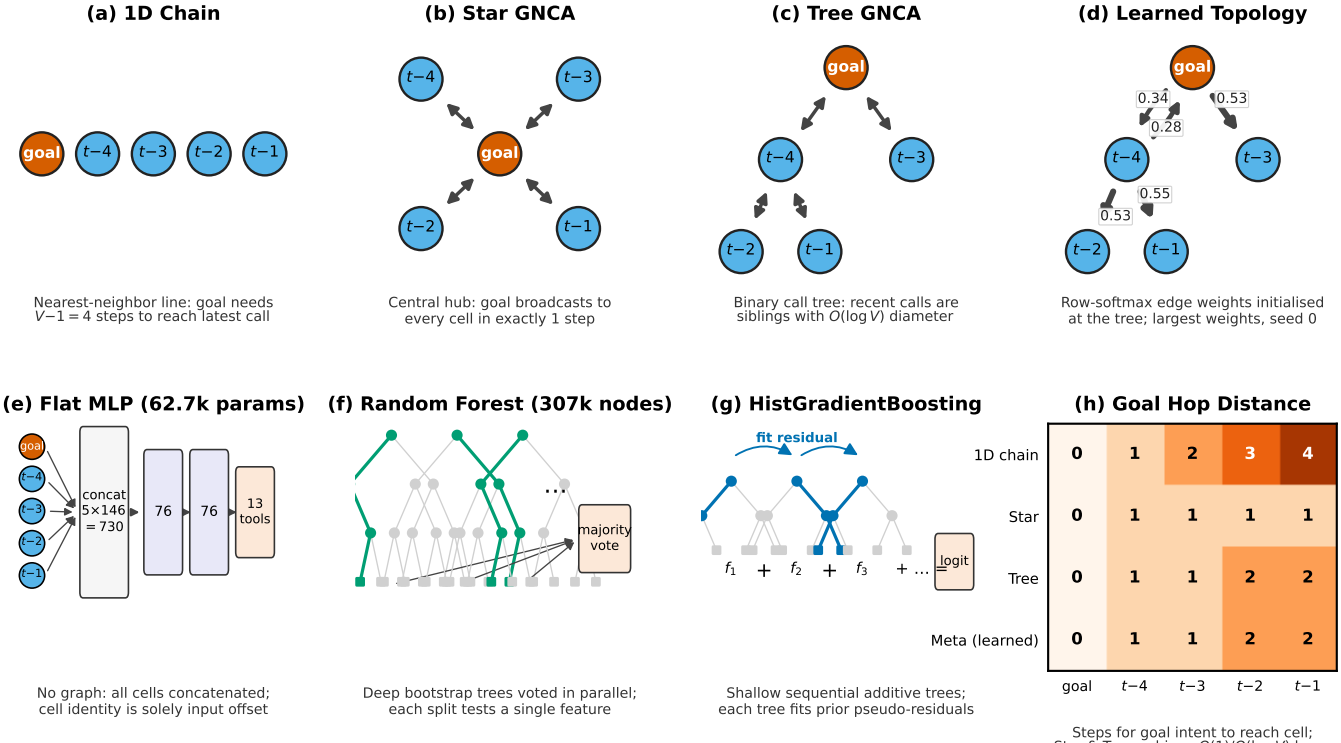

(a) 1D Chain
goal
t−4
t−3
t−2
t−1
Nearest-neighbor line: goal needs V−1 = 4 steps to reach latest call
(b) Star GNCA
goal
Central hub: goal broadcasts to every cell in exactly 1 step
(c) Tree GNCA
Binary call tree: recent calls are siblings with O(log V) diameter
(d) Learned Topology
0.34
0.53
0.28
0.55
0.53
Row-softmax edge weights initialised at the tree; largest weights, seed 0
(e) Flat MLP (62.7k params)
concat 5×146 = 730
76
76
13 tools
No graph: all cells concatenated; cell identity is solely input offset
(f) Random Forest (307k nodes)
majority vote
Deep bootstrap trees voted in parallel; each split tests a single feature
(g) HistGradientBoosting
fit residual
logit
Shallow sequential additive trees; each tree fits prior pseudo-residuals
(h) Goal Hop Distance
1D chain 0 1 2 3 4
Star 0 1 1 1 1
Tree 0 1 1 2 2
Meta (learned) 0 1 1 2 2
goal t−4 t−3 t−2 t−1
Steps for goal intent to reach cell; Star & Tree achieve O(1)/O(log V) hops